
Impossible Biomimetic Synthesis: AI Reasoning in Engineering Design

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Abstract

We designed a hierarchical, constrained multi-agent synthesis framework to study how a language model reasons when tasked with engineering design. The system instructed a language model to synthesize a hydraulic rock breaker with a pistol shrimp. Given the deep incompatibility between these systems, any coherent synthesis of the two was extremely unlikely. This deliberate adversarial framing however created conditions for observing where model reasoning succeeds and where it degrades. The framework decomposed mechanical systems across six levels of abstraction, enforced provenance tracking against source specifications, and validated outputs against 22 physics and structural constraints. The system produced seven discrete designs of varying technical complexity. We observed design decisions reflecting genuine synthesis and creative problem-solving, but also spurious rationalizations and difficulty sustaining multi-step inference and managing long-range dependencies. Overall, our results suggest that language models can effectively navigate constrained design spaces, although nothing in our findings indicates they can perform the radical restructuring and reconceptualization underlying paradigm-shattering breakthroughs. AI's most productive contribution to engineering may be in accelerating the systematic exploration and incremental optimization that support the engineering process, while the breakout insights that drive profound advances still exceed the capabilities of current architectures.

1. Introduction

Amid the hyperbole and messianic claims, one application of AI continues to loom large and command serious attention – that of scientific discovery. While some cast language models as glorified token prediction, others view them as sophisticated and versatile learning architectures. Hardliners in this latter camp argue that all cumulative intellectual progress is merely scaffolding for the arrival of more advanced computational machinery.

Scientific discovery is elusive and amorphous, and efforts to systematically study it are fraught. But the same creative energy underlying discovery also animates a different and more empirically constrained form of invention: engineering design.

Expanding new frontiers places ongoing demands on the engineer. Although this creative process resists formulaic description, it is mediated by imagination, intuition,

and experience. This raises the question: can creative technical work be reduced to AI-orchestrated algorithmic search?

As a test of this proposition, we designed and implemented a system of technical specification, product design, and constraint and requirement validation – all fully executed by a language model. This was structured as a hierarchical, constrained multi-agent synthesis framework with LLM-based provenance and physics validation. In our architecture, a language model iteratively moves across different levels of design abstraction, selects components from a closed library of primitives, and satisfies provenance tracking and physics constraints. Grounding the model output in these requirements enforces coherence, restricts erroneous invention, and prevents violations of the laws of physics.

Our experiment was framed as a synthesis task: we instructed a language model to synthesize an existing industrial tool (hydraulic rock breaker) with a biological organism (pistol shrimp) to generate a novel hybrid mechanical design. This specific pairing was intentional. Although both possess a striking mechanism, the rock breaker and pistol shrimp use disparate energy systems (percussion vs cavitation), work in different mediums (air vs water), and operate at energy scales separated by orders of magnitude (kilowatts vs milliwatts). This adversarial pairing was almost certain to result in failure – the AI faced an impossible task of reconciling fundamentally incompatible systems.

Counterintuitively, this adverse pairing was advantageous for our work. Language models, when forming hypotheses, frequently resort to handwaving or superficially plausible explanations that are ill-conceived or incoherent. An adversarial pairing combined with robust validation offers a useful stress test. As the model strains to find unorthodox solutions, we gain visibility into parts of the pipeline where performance degrades. Although we know in advance that the model will fail, knowing exactly how it fails is instructive.

Our results yielded several findings: residual spurious reasoning despite constrained validation, occasional moments of sharply creative problem-solving, and weaknesses in managing long-term dependencies and multi-step inference. These findings inform a broader discussion on the nature of creativity in technical disciplines and the role of AI architectures in supporting engineering and scientific discovery.

2. Methodology

2.1 Case Selection

Biomimetic design – leveraging biological principles in building mechanical objects – has long influenced advances in robotics and engineering (Bar-Cohen, 2011). This motivates the pairing of an industrial tool with a biological organism and enables drawing on natural selection as a source of novelty in mechanical design.

The specific pairing of the hydraulic rock breaker and pistol shrimp was driven both by the sharp structural asymmetry between the two entities and by partial superficial similarity. Both systems (we treat the pistol shrimp as a mechanical object for the purposes of our study) deliver a powerful expulsion of force – the rock breaker through pressurized percussion impact and the pistol shrimp through a cavitation shockwave generated by a snapping claw.

Each however operates according to distinct principles. The pistol shrimp leverages power amplification – it accumulates and stores energy slowly, then releases it instantaneously. The resulting explosive water jet (accelerated to 25 m/s) forms a vaporized bubble which collapses and lethally stuns the shrimp’s prey. The rock breaker on the other hand delivers high force and continuous power through pressurized liquid. Yet it lacks the shrimp’s energy accumulation mechanism and is ill-suited to explosive bursts of energy.

This adversarial pairing, combined with strict provenance and physics constraints, created the conditions for our experimental framework.

2.2 Experimental Framework

The source specifications for both systems were developed according to a three-stage LLM-facilitated workflow: identifying authoritative sources, extracting technical parameters with accompanying citations, and structuring data according to the Engineering Decomposition Framework (EDF). This framework decomposes mechanical systems across six hierarchical levels (Table 1) progressing from high-level governing concepts (L0) to precise quantitative parameters (L5).

Table 1: EDF Abstraction Hierarchy

Level	Focus	Example
L0	Governing concept and operating principle	Hydraulic-powered cavitation mechanism for rock breaking
L1	Energy flow sequence from input to output	Hydraulic input → elastic storage → latch release → fluid acceleration → cavitation
L2	Domain primitives from closed lists	A2+A3 (hydraulic+elastic), B2 (latch), C1 (linear), D2 (fluid), E2 (sealed), F2 (electronic)
L3	Specific mechanisms implementing primitives	Latch-release valve, pressure accumulator, linear guide rail
L4	Component parts list and materials	Steel housing, piston assembly, hydraulic lines, tendon cables
L5	Quantitative parameters with units	210 bar pressure, 750 J stored energy, 25 m/s jet velocity, 40 ms cycle time

Table 2: Functional Domains

Domain	Code	Primitive
A: Energy Source	A1 / A2 / A3	Metabolic · Hydraulic · Pneumatic/Elastic
B: Energy Release	B1 / B2	Continuous Cycle · Instantaneous Latch
C: Kinematic Constraint	C1 / C2 / C4	Linear Guide · Revolute Joint · Linkage
D: Working Interface	D1 / D2	Percussion Impact · Fluid Acceleration
E: Sealing/Containment	E2 / E3	Dynamic Seal · Ambient/Open
F: Control Interface	F1 / F2 / F3	Mechanical Pilot · Electronic Discrete · Neural/Biological

At L2, designs must select primitives from six functional domains: Energy Source (A), Energy Release (B), Kinematic Constraint (C), Force Delivery (D), Sealing/Containment (E), and Control Interface (F). Domain codes are drawn from a closed list (e.g., A2=hydraulic, B2=instantaneous latch, D2=fluid acceleration) to enforce methodological consistency and prevent invalid combinations.

Every element in a specification carries a provenance tag indicating its origin (Table 3). The synthesis agent operates under defined constraints - it must ground its outputs in the source specifications. Novel elements are permitted only as a last resort and are subject to explicit justification. This prevents unverified hallucination of features derived from the model's general knowledge of pistol shrimp and hydraulic systems.

Table 3: Provenance Tag System

Tag	Meaning	Usage Priority	Justification Required
[FROM-A]	Direct transfer from Spec A (Pistol Shrimp)	Highest	No
[FROM-B]	Direct transfer from Spec B (Rock Breaker)	Highest	No
[ADAPT-A]	Modified from Spec A with stated changes	Medium	Modification must be explicit
[ADAPT-B]	Modified from Spec B with stated changes	Medium	Modification must be explicit
[SYNTH]	Novel element from neither source	Lowest (last resort)	Must explain why sources inadequate

Provenance claims undergo active verification through a "mini-RAG" system. The structural validator maintains access to the source specifications as YAML files and

searches these documents when encountering provenance tags. A claim of "[FROM-A] plunger diameter 25mm" triggers a search of Spec A for this technical specification to ensure strict grounding in the curated specifications. Designs exceeding 30% SYNTH content are flagged for excessive invention rather than synthesis.

2.3 Multi-Agent Architecture

The system architecture employs three LLM-based agents - a synthesis agent, a physics validator agent, and a structural validator agent - coordinated by Python orchestration code. This multi-agent architecture combines LLM reasoning with control flow logic to ensure consistent validation and methodological rigor.

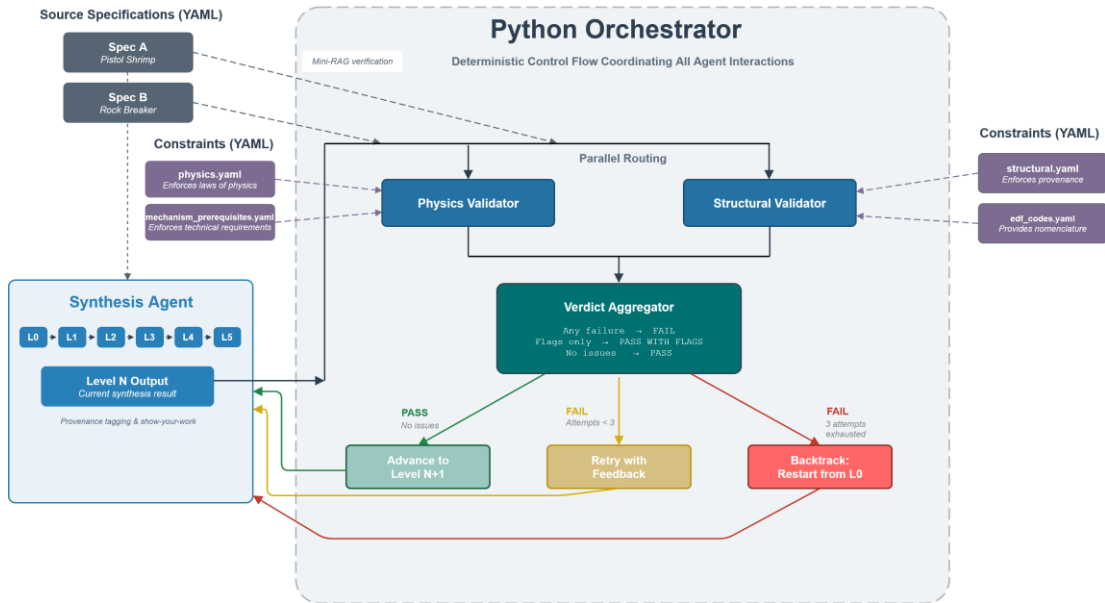


Figure 1: System Architecture and Information Flow

2.3.1 Synthesis Agent

The synthesis agent operates under defined constraints enforced through prompt instructions. It is instructed to prioritize information from Specification A (pistol shrimp) and Specification B (rock breaker) and to resort to novel synthesis only when neither source can serve the required function. It must tag every element with provenance codes indicating its origin. The agent receives validated output from previous synthesis stages as well as feedback from failed validation attempts. It operates without knowledge of retry limits or orchestration logic.

The model temperature was set to 0.4 to enable creative exploration of the design space while maintaining grounding in source material. The agent is required to "show its work" for quantitative claims at L5 and to provide explicit calculations that validators can verify.

2.3.2 Physics Validator

The physics validator uses an independent calculation protocol. Rather than simply verifying that calculations are shown, the validator is instructed to extract parameter values from the synthesis specification, perform arithmetic independently, and compare its results to the agent's claims. The validator operates at a temperature of 0.1 to promote consistent outputs across executions. It enforces thresholds defined in YAML configuration files which serve as a source-grounded baseline.

2.3.3 Structural Validator

The structural validator works to prevent source hallucination through active document verification. It accesses Source Specifications A and B (YAML files) and searches these documents to verify provenance claims. If a discrepancy is found, a failure outcome is produced with feedback specifying the divergence. This "mini-RAG" system enforces grounding in source-based specifications and prevents reliance on the agent's general knowledge about pistol shrimp or hydraulic systems.

The validator also enforces parsimony by requiring explicit justification for every [SYNTH] element. The agent must explain why neither source specification can serve the required function. Vague justifications such as "for better performance" or "to optimize design" trigger immediate failure. Valid justifications must specifically explain why each source is inadequate in supporting the proposed function. The model temperature was set to 0.1 to ensure consistent enforcement of structural constraints across validation runs.

2.3.4 Orchestration and Control Flow

Python orchestration code coordinates agent interactions and manages synthesis progression. The output of the synthesis agent routes to both validators in parallel for independent evaluation. The verdict aggregator combines their outputs using fixed rules: any constraint failure yields FAIL, flags without failures yield PASS_WITH_FLAGS, and outputs without any issues yield PASS. The orchestrator has no LLM component. All decision logic is deterministic code to ensure consistent behavior.

2.4 Constraint System

The physics and structural validation agents together employ 22 constraints across two tiers. Universal physics constraints (physics.yaml) apply to all designs regardless of the mechanisms chosen, whereas mechanism-specific prerequisites (mechanism_prerequisites.yaml) are invoked only when specific mechanisms require additional validation against source system parameters.

Table 4: Physics Constraints

ID	Name	Formula/Check	Applies	Example Violation
P1	Energy Conservation	$E_{\text{available}} \geq 0.5mv^2$	L5	Design 5: 400 J cable capacity, system needs 750 J
P2	Temporal Feasibility	$T_{\text{cycle}} \geq T_{\text{charge}} + T_{\text{fire}} + T_{\text{reset}}$	L5	Claimed 40 ms cycle, phases sum to 48 ms
P3	Mechanism Prerequisites	Alignment with requirements in mechanism_prerequisites.yaml	L5	Cavitation at 18 m/s (threshold: ≥ 25 m/s)
P4	Dimensional Consistency	Units consistent across calculations	L5	Mixing meters with millimeters in force calculation
P5	Power Adequacy	$P_{\text{source}} \geq E_{\text{cycle}} \times f_{\text{cycle}}$	L5	Design 2: 21 kW input, claims 80 kW sustained output
P6	No Magic Physics	Obeys known physics laws	All	Perpetual motion, energy from nothing
P7	Energy Continuity	Output connects to next input	L1	Gap in energy flow chain
P8	Cycle Closure	Returns to initial state	L1	One-shot sequence, can't repeat
P9	Rate Limiter Identified	Slowest phase identified (flag)	L1	Bottleneck not specified
P10	Kinematic Continuity	Force paths physically connected	L3	Force "teleportation" without transmission path
P11	Material Reality	Real industrial materials only	L4	Fictional "unobtainium alloy"
P12	Efficiency Plausibility	Flag if claimed efficiency $> 95\%$	L5	97% efficiency claim (flag for review)

Table 5: Structural Constraints

ID	Name	Check	Focus
S1	Closed-List Compliance	Valid EDF codes, no prohibited combinations	B1+B2 mutually exclusive
S2	Provenance Accuracy	Mini-RAG verification against source specs	[FROM-A] claims must exist in Spec A

S3	Domain Coverage	All six domains (A-F) addressed	Complete functional specification
S4	Vertical Traceability Down	Elements at N trace to parent at N-1	No orphan elements
S5	Vertical Traceability Up	Elements at N-1 realized by children at N	No dangling requirements
S6	Primitive Coverage	L2 primitives map to ≥ 1 L3 mechanism	Complete implementation
S7	Mechanism Coverage	L3 mechanisms decompose to L4 components	Complete parts list
S8	Parameter Completeness	Key parameters specified (flag)	Encourage completeness
S9	SYNTH Justification Required	[SYNTH] elements need necessity explanation	Anti-gaming enforcement
T1	Parsimony Compliance	[SYNTH] usage $\leq 30\%$ of elements (flag)	Statistical source utilization check

Structural constraints enforce EDF methodology compliance and prevent source hallucination. S2 performs the critical mini-RAG verification, while S9 serves as the primary source grounding mechanism by requiring high-quality justifications for novel synthesis. T1 monitors source composition by flagging designs that rely excessively on invention rather than source-based synthesis.

Table 6: Constraint Activation by Level

Level	Active Physics Constraints	Active Structural Constraints
L0	P6	S2, S9, T1
L1	P6, P7, P8, P9	S2, S4, S5, S9, T1
L2	P6	S1, S2, S3, S4, S5, S9, T1
L3	P6, P10	S2, S4, S5, S6, S9, T1
L4	P6, P11	S2, S4, S5, S7, S9, T1
L5	P1-P6, P12	S2, S4, S5, S8, S9, T1

This staged constraint application ensures conceptual validity is established early in the pipeline (L0-L2) before expending computational effort to determine quantitative specifications (L5). The system invokes quantitative constraints (P1, P2, P3, P5) only at L5.

2.5 Iterative Descent and Validation Protocol

The synthesis operates sequentially with validation at each level of the framework. After the synthesis agent generates content for level N, the orchestrator routes its output to both validators in parallel for evaluation. Each validator returns a structured JSON response with the constraint results.

Each level permits three validation attempts before triggering backtracking. The validators provide feedback on failed attempts specifying which constraints were violated, the threshold values versus actual values for quantitative constraints, and specific remediation options.

If all three attempts at level N fail validation, the orchestrator abandons the current synthesis trajectory and restarts synthesis from L0. The system saves the failed synthesis trajectory's completed levels as a partial design attempt and then extracts a failure signature comprising constraint violation counts and the L2 domain primitive combination (e.g., "A2+A3|B2|C1|D1+D2|E2|F2"). This signature is injected as a negative constraint into the new L0 context instructing the synthesis agent to explore a fundamentally different design approach and avoid re-exploring primitive combinations that led to irreconcilable conflicts.

3. Results

3.1 Overview

The synthesis framework generated seven discrete designs across its full execution. No design passed L5 parameter specifications. Three designs (Designs 2, 4, and 6) were terminated before reaching L5. The remaining four (Designs 1, 3, 5, and 7) reached L5 but failed to pass validation within the permitted retry attempts. Design 7 came closest to completion and accumulated only 8 total violations across its entire run. Design 1 accumulated the most at 95. Table 7 provides an overview of all seven designs.

Table 7: Design Overview

Design	Fingerprint	Completed Levels	Total Violations	Total Tags	Status
1	A2+A3 / B2 B1 / C1 C2+C4 / D2 D1 / E2+E3 / F2	L0-L4	95	146	Failed L5 (9 attempts)
2	A2+A3 / B2 / C1 / D2 / E2 / F2	L0-L3	17	56	Terminated at L4
3	A2+A3 / B2 / C1 / D1 / E2 / F2	L0-L4	80	216	Failed L5 (9 attempts)
4	A2+A3 / B2 / C1+C4 / D2 / E2 / F2	L0-L2	30	79	Failed at L3
5	A2+A3 / B2 / C1 / D1+D2 / E2 / F2	L0-L4	91	153	Failed L5 (9 attempts)
6	A2+A3 / B2 / C1 / D2 / E2 / F2	L0-L3	9	115	Terminated at L4

7	A2+A3 / B2 / C1+C2 / D2 / E2 / F2	L0-L4	8	144	Failed L5 (9 attempts)
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Across their domain fingerprints, all seven designs shared the following selections: A2+A3 (hydraulic + elastic energy source), B2 (instantaneous latch release), E2 (sealed containment), and F2 (electronic control). The designs diverged primarily in their kinematic constraint (C) and force delivery (D) selections. Design 1 was the only design to use mode-dependent primitives denoted by the "|" operator. It was also the only design to employ E2+E3 hybrid sealing.

Table 8: Constraint Violations by Type

Design	P1	P2	P3	P4	P5	P6	P10	S2	S7	S9	Total
1	27	13	9	2	8	1	-	22	-	13	95
2	-	-	-	-	-	-	-	9	2	6	17
3	8	-	15	-	3	2	-	38	-	14	80
4	-	-	3	-	-	-	2	13	-	12	30
5	16	15	8	-	9	2	-	25	-	16	91
6	-	-	2	-	-	1	-	4	-	2	9
7	1	-	-	-	1	-	-	4	-	2	8

Energy conservation (P1) was the most frequent physics violation. Design 1 recorded 27 P1 violations - the highest single-constraint count of any design. Provenance accuracy (S2) was the most frequent structural violation with Design 3 recording 38 - the highest S2 count across all designs. SYNTH justification (S9) failures were distributed broadly with Designs 5 and 3 recording the highest counts (16 and 14 respectively). Temporal feasibility (P2) violations appeared only in designs attempting dual force delivery: Designs 1 (13) and 5 (15). Kinematic continuity (P10) violations were unique to Design 4.

Table 9: Constraint Violations by Level

Design	L0	L1	L2	L3	L4	L5	Total
1	0	0	0	0	7	88	95
2	0	4	0	1	12	-	17
3	0	5	0	0	10	65	80
4	0	0	4	26	-	-	30
5	0	0	0	0	1	90	91
6	0	1	0	3	5	-	9
7	1	0	0	0	0	7	8

Violations were overwhelmingly concentrated at L5 for the four designs that reached that level. Design 5 accumulated 90 of its 91 total violations at L5 after passing L0 through L3 relatively cleanly. Design 1 followed a similar constraint violation

concentration with 88 of 95 violations at L5. Design 4 presented the sharpest contrast with 26 of its 30 violations occurring at L3 where mechanism specification proved irreconcilable with its C1+C4 kinematic selection. For Designs 2 and 6 (both terminated at L4) violations were distributed more gradually across L1, L3, and L4.

Table 10: Provenance Distribution

Design	FROM-A	FROM-B	ADAPT-A	ADAPT-B	SYNTH	Total	SYNTH %
1	8 (5.5%)	46 (31.5%)	24 (16.4%)	3 (2.1%)	65 (44.5%)	146	44.5%
2	11 (19.6%)	11 (19.6%)	12 (21.4%)	10 (17.9%)	12 (21.4%)	56	21.4%
3	18 (8.3%)	134 (62.0%)	38 (17.6%)	5 (2.3%)	21 (9.7%)	216	9.7%
4	20 (25.3%)	22 (27.8%)	15 (19.0%)	6 (7.6%)	16 (20.3%)	79	20.3%
5	30 (19.6%)	44 (28.8%)	25 (16.3%)	20 (13.1%)	34 (22.2%)	153	22.2%
6	26 (22.6%)	26 (22.6%)	29 (25.2%)	5 (4.3%)	29 (25.2%)	115	25.2%
7	22 (15.3%)	40 (27.8%)	36 (25.0%)	13 (9.0%)	33 (22.9%)	144	22.9%

Design 3 drew most heavily from a single source with 62.0% of its tags originating directly from the rock breaker specification - consistent with its decision to drop cavitation and commit to percussion. Design 2 had the most balanced source distribution with near-equal representation across all five tag categories. Design 1 had the highest SYNTH element proportion at 44.5% which was well above the 30% parsimony threshold. This was driven by the mode-switching components required by its dual-mode design. Design 7 maintained a moderate SYNTH rate of 22.9% while achieving the strongest overall progression.

Design 1: Dual-Mode Impact System

Design 1 attempted the most ambitious synthesis: a system capable of switching between cavitation and percussion. Rather than combining both force delivery mechanisms simultaneously, it treated them as discrete operational modes with mode-dependent domain primitives. This required hybrid sealing (E2+E3) to support operation in both fluid-coupled and dry environments. Design 1 was the only design to employ mode-dependent notation and the only design with hybrid sealing.

The design passed L0 through L3 cleanly before accumulating seven violations at L4 as mode-switching hardware introduced elements not traceable to either source. L5 was deeply problematic producing 88 violations across nine failed attempts. Energy conservation violations (P1) dominated with 27 occurrences - the dual-mode design could not satisfy energy budgets for both operational paths simultaneously. The

provenance profile was heavily skewed toward novel synthesis with SYNTH at 44.5%. This was the highest of any design and well above the 30% parsimony threshold.

Design 2: Pure Cavitation (Minimalist)

Design 2 dropped percussion entirely and committed to pure cavitation. Its fingerprint was the simplest of any design with a single primitive per domain and no combinations or alternatives. It shares its fingerprint with Design 6. Both attempted to scale the pistol shrimp's cavitation mechanism to industrial dimensions and both were terminated at L4.

Unlike most designs, Design 2 showed violations early with failures appearing at L1, L3, and L4 rather than clustering at a single level. It was the only design to incur S7 (mechanism coverage) violations. Its provenance distribution was the most balanced of any design with near-equal representation across all five tag categories and a SYNTH rate of 21.4%.

Design 3: Elastic Impact (Pure Percussion)

Design 3 took the opposite approach relative to Design 2. It dropped cavitation entirely and committed to pure percussion (D1). It applied the pistol shrimp's power amplification principle - elastic storage with an instantaneous latch release (B2) - to the rock breaker's percussive force delivery mode. Its L0 reasoning explicitly referenced prior design failures. This B2-D1 pairing was unique among all seven designs.

Provenance issues emerged immediately at L1 as the cross-domain abstraction and transfer of principle proved too difficult to ground. L4 accumulated 10 S2 violations despite the highest tag count of any design at that level (96 tags). L5 produced 65 violations across nine attempts with mechanism prerequisites (P3) as the leading physics failure at 15 occurrences. The provenance profile was dominated by FROM-B at 62.0% which was by far the highest single-source reliance across all designs. This was consistent with its commitment to the rock breaker's percussion mode. SYNTH was the lowest of any design at 9.7%.

Design 4: Cavitation with Linkage Kinematics

Design 4 added linkage mechanics (C4) to the cavitation-focused approach selecting C1+C4 to leverage the pistol shrimp's linkage-based power amplification alongside linear guidance. It was the shortest-lived design and completed only L0 through L2 before failing at L3.

Performance at L3 was deeply problematic: 26 of its 30 total violations occurred at L3 across three failed attempts. The C1+C4 combination created motion discontinuities that the validators rejected and Design 4 was the only design to incur P10 (kinematic continuity) violations. SYNTH justification failures (S9) dominated at 12 occurrences as the validators rejected the invented linkage mechanisms as unjustified. The provenance distribution was relatively balanced with the highest FROM-A count at L0

(6 tags) of any design indicating strong initial grounding in the pistol shrimp specification.

Design 5: Cavitation-Enhanced Percussion

Design 5 combined both force delivery modes (D1+D2) with simple linear kinematics (C1) and attempted to use cavitation and percussion together rather than as discrete alternatives. This was a simpler implementation of the dual-mode concept than Design 1 and without mode switching.

The design passed L0 through L3 without a single violation before producing 90 of its 91 total violations at L5. No other design exhibited such an abrupt shift. Unlike other high-violation designs, failures were distributed broadly across constraint types with no single dominant mode. Its temporal feasibility (P2) count of 15 was the highest of any design and the combined force delivery commitment created timing conflicts even without explicit mode switching. Design 5 had the highest ADAPT-B count (20) of any design which reflected substantial adaptation of rock breaker components.

Design 6: Pure Cavitation (Second Attempt)

Design 6 shared Design 2's fingerprint. It represented a second attempt at the minimalist pure cavitation approach. It completed L0 through L3 and was terminated at L4. Its total of 9 violations was the second lowest of any design.

Despite identical domain primitive selections, the two cavitation designs diverged in specification and failure profile. Design 6 had higher total tags (115 vs. 56) and a higher SYNTH proportion (25.2% vs. 21.4%). At L3, SYNTH dramatically rose to 19 tags (37% of the level) - the highest L3 SYNTH count of any design. Design 2 struggled with distributed violations from L1 onward, whereas Design 6 progressed more cleanly before encountering difficulties at L3.

Design 7: Cavitation with Revolute Joint Kinematics

Design 7 selected C1+C2 (linear guide + revolute joint) for cavitation-only force delivery (D2) using the revolute joint's rotational motion to generate cavitation. It completed L0 through L4 and reached L5 with only 8 total violations - far fewer than any other design.

A single S2 violation appeared at L0 followed by clean passage through L1 to L4. L5 produced just 7 violations. Design 7 incurred just two physics violations across its entire run and was the only design to reach L5 with zero P2 (temporal feasibility) and zero P3 (mechanism prerequisites) violations. At L2, SYNTH was minimal (1 of 17 tags) - the lowest of any design - indicating near-complete source grounding at the domain primitive level. At L1, it had the highest FROM-A count (11 tags) of any design reflecting strong direct transfer of pistol shrimp mechanisms into the energy flow specification.

4. Discussion and Analysis

Our system explored discrete designs and surfaced a range of engineering principles and technical patterns. But the experimental scope was deliberately limited - we designed a hierarchical, constrained synthesis with an adversarial pairing featuring irreconcilable systems. Consequently, no significant engineering outcome was expected to emerge. The primary value of these results however is in what they reveal about the language model’s reasoning capabilities and the feasibility of using AI to orchestrate engineering design.

We examine specific design decisions below - both failures and successes - before turning to broader questions about AI’s capacity for creative engineering and scientific discovery.

4.1 Spurious Reasoning

One recurring and conspicuous deficiency in our results was the language model’s tendency to generate plausible-sounding rationales disconnected from engineering rigor and reality. For instance, the L0 reasoning of Design 1 states that a dual-mode approach allows the system to “leverage the shrimp’s precision and energy efficiency for delicate operations”. This led it to enlist the shrimp’s “high-velocity fluid mechanism” for “precision applications requiring cavitation effects”.

This however betrays a lack of semantic understanding of these concepts. Cavitation is a fundamentally violent and chaotic phenomenon. The collapse of the pistol shrimp’s cavitation bubble generates shockwaves and temperatures briefly exceeding 4700°C. Arguing that this would benefit the synthesized object’s ability to undertake “precision” or “delicate” operations is clearly implausible and erroneous.

This phenomenon is likely an artifact of the underlying mechanics of language models. As statistical prediction engines, LLMs are prone to spurious associations and hallucinations, and often prioritize semantic plausibility over factual or physical grounding (Ji et al., 2023). It appears that the language model is conflating the small-scale properties of the shrimp’s cavitation mechanism with ideas of precision.

This points to a limitation of our experimental design. Namely, while we defined constraints to ensure provenance tracking, EDF compliance, and consistency with physics rules, we did not apply a control to validate semantic understanding.

Given the demonstrable value of self-critique and refinement in enhancing language model performance (Madaan et al., 2023), our work would have benefited from the introduction of a ‘critic’ or ‘red team’ agent. Such an agent would have been instructed to critically review the outputs of the synthesis at earlier, more conceptual stages. The critical feedback provided would have helped mitigate erroneous reasoning during idea incubation and would have anchored the generated ideas in a more rigorous foundation.

Admittedly, this flawed line of reasoning did not exert any real influence on further synthesis progression - the design was ultimately undermined by its difficulty satisfying structural validation constraints. Regardless, we must be mindful of such missteps. Early divergences, errors, and misinterpretations, however small, can propagate through the pipeline and expand in magnitude as they amplify across decision nodes.

4.2 Flawed Causal Inference and Long-Term Dependencies

While our validation constraints helped guard against outright errors, the model still made subtle reasoning mistakes related to multi-step causal inference. We examine three prominent examples:

- **Design 5: Component undersized for specification.** At L1, the system committed to storing 750 J of elastic energy per cycle. At L4, when specifying the component storing this energy, the agent described a "12mm diameter high-strength steel cable with 100mm active length to achieve 400 J storage capacity." The component's maximum capacity is 400 J, but the system requires 750 J. The system is 350 J short of its own specification.
- **Design 2: Claimed performance exceeds available power.** At L0, the system promises "1-10 kJ per pulse." L1 repeats this claim while also specifying a 21 kW hydraulic input and 32-121 ms cycle times. But energy per pulse depends on the amount of time the system has to accumulate energy from a fixed-power source before firing. At 21 kW, accumulating 1 kJ requires 48 ms and accumulating 10 kJ requires 476 ms. The specified cycle time range of 32-121 ms only allows 672-2,541 J per pulse. The 10 kJ upper bound is impossible at any of the specified cycle rates as it would require nearly four times the maximum cycle time. Even the 1 kJ lower bound is only achievable at cycle times above 48 ms which excludes the faster end of the claimed operating range. The agent could not effectively navigate the trade-off between high energy per pulse and high cycling frequency.
- **Design 6: Biological parameter copied without scaling.** At L1, the latch holding force is specified as 50 N - the exact value from the pistol shrimp's biological specification. At L3, the elastic storage system is described as "25mm diameter" tendons charged by "150 bar" hydraulic pressure. Hydraulic pressure at that level on any reasonable piston area produces forces in the kilonewtons. Design 3, facing the same problem, correctly scaled its latch to hold 10,500 N. Design 6 however did not make the corresponding adjustment. A 50 N latch cannot restrain kilonewton-scale charging forces - the latch would be released the instant hydraulic pressure was applied. The agent transferred a source parameter that required scaling but failed to perform the scaling.

These three failures share a common deficit - the model reasons locally rather than globally. Design 5 specified a component without checking it against an energy

commitment made two levels earlier. Design 2 selected three independently reasonable parameters (energy per pulse, cycle time, and power input) without recognizing their joint incompatibility. Design 6 transferred a parameter without performing the appropriate scaling. In each case, the model produced specifications that were locally but not globally coherent.

Validation and constraints can mitigate these, but they cannot permanently overcome the structural limitations of model reasoning. Language models have demonstrated considerable fluency on well-defined problems boosted by techniques like chain-of-thought prompting (Wei et al., 2022), but this fluency degrades as the model must maintain consistency across multiple levels of transformation. This challenge compounds with system complexity and is consistent with findings that language model performance degrades significantly when critical information stretches across long context windows (Liu et al., 2024). Strategies like embedding explicit verification into prompt instructions become harder to enforce as the number of long-term dependencies grows.

4.3 AI's Capacity for Creative Design

The model's reasoning however was not always disjointed. Three designs in particular exhibited moments of genuine creative problem-solving.

Design 1 – Dual-Mode

In Design 1, the model envisaged a dual-mode system sequentially switching between cavitation and percussion. It saw this as synergistic and defended it with a sound rationale: percussion breaks up hard rock first, then cavitation attacks the fractures created by the initial impact. This represents an effort at genuine synthesis and has the trappings of an emergent property - a capability that exists in neither source system alone but emerges from their combination.

More striking was a form of structural creativity not seen in any of the other designs. The prompt instructions indicated that functional domains could be combined (barring B1+B2) but provided no explicit conception of alternating or switching between modes. Design 1 independently invented a new notation - the "|" character - to denote a dual-mode system signifying sequential operation in "either mode." The model effectively extended the framework's expressiveness to accommodate its proposed design. This came however at a considerable cost as 44% of all provenance tags associated with Design 1 were tagged SYNTH.

Design 3 – Concept Abstraction

Design 3 performed a subtler feat of creative reasoning: it extracted a principle from one source while discarding the mechanism that embodied it. This allowed it to achieve a more robust synthesis without incurring the constraint violations other designs contended with. Understanding how it did this calls for some explanation of the energy systems of the two sources.

The pistol shrimp's cavitation mechanism has three components: slow energy accumulation, instantaneous release, and forced delivery via cavitation. The rock breaker's striking mechanism however is governed by a different dynamic – hydraulic power drives a continuous cycle of pressurized fluid which delivers percussion impact.

Design 3 abstracted the principle of instantaneous release (B2) and routed it to the rock breaker's delivery mode (D1) forgoing the use of cavitation and fluid acceleration (D2). This B2-D1 pairing was unique among all the designs. It allowed Design 3 to escape a profound challenge - scaling fluid acceleration to industrial dimensions. At an industrial scale, accelerating fluid volume to extreme velocities (e.g. 25+m/s) has a disproportionate energy cost and this repeatedly created physics constraint violations. Design 3 seized the power amplification benefit without having to solve cavitation scaling.

Although Design 3 had a relatively low number of physics violations, it had the highest number of provenance violations due to this novel combination. The structural validator treated instantaneous latch release (B2) and cavitation (D2) as tightly coupled. This reflected their pairing in the pistol shrimp's biological specification. Yet by retaining B2 but routing it to percussion (D1), Design 3 was repeatedly penalized by provenance constraints that expected B2 to accompany D2.

Design 4 – Parallel Charging

Design 4 creatively addressed a core timing constraint. The pistol shrimp's latch-release mechanism requires 320ms to recharge between cycles. The rock breaker however operates at 25 Hz which translates to strikes every 40ms. A single shrimp style latch-release mechanism cannot recharge fast enough to sustain this rate. Design 4 proposed using eight energy accumulators in parallel, independently arriving at a time-tested engineering strategy seen in systems like multi-cylinder engines.

Design 4 however suffered from a deficiency not exhibited by any other design – the quality of its SYNTH justifications rapidly deteriorated. The framework requires explicit justification for every SYNTH component. While other designs provided robust justifications, Design 4's declined substantially in quality and specificity. This is presumably why it was terminated earliest and failed to pass L3.

Why did performance deteriorate mid-execution? A definitive explanation is impossible given that LLMs are stochastic and their outputs can fluctuate unpredictably. Our decision to use a relatively high model temperature (0.4) for the synthesis agent may have exacerbated this. Other contributing factors could include context window pressure or the difficulty of reasoning coherently about eight parallel accumulators simultaneously.

4.4 Evaluating Creative Output

Assessing and opining on creativity requires some working criterion. Rather than build a philosophically fraught formal definition, we apply a modest standard: a

design choice qualifies as creative when it is surprising, impressive, or when it produces emergent capabilities not present in either source system. This can be distilled into an even more straightforward test: if a human engineer had made this decision, would it have elicited a similar reaction?

This rubric however must be placed in context. The rock breaker has been intensively studied, engineered, and refined over decades. Its present form reflects cumulative optimization across performance, cost, energy efficiency, manufacturability, and operational complexity. Our synthesis was therefore not likely to improve on this in any way. Even a coherent hybrid construction was improbable given the selected adversarial pairing with its irreconcilable energy systems.

Yet against this backdrop the designs were not just lackluster write-offs. While they produced no breakthroughs, they demonstrated an ability to effectively explore the constrained design space. Even more striking, they showed genuine synthesis that went beyond uncritical recombination.

4.5 AI and the Structure of Engineering Progress

The most ambitious test of creative engineering design must grapple with the most significant driver of engineering progress: groundbreaking breakthroughs. Our system did not produce such breakthroughs. But the question extends beyond our experiment - could an AI system be capable of such innovation? There are reasons to be skeptical.

The history of the engine is instructive. With some simplification, we can view progress in engine design as advancing along two tracks: long stretches of incremental refinement punctuated by moments of radical restructuring and redesign. Consider Watt's introduction of a separate condenser, the double-acting cylinder, or the steam turbine. These were not minor optimizations within existing designs but instead represented fundamental reconsiderations of what an engine could be. The separate condenser required recognizing that heating and cooling the same cylinder was not a problem requiring superior permutations but a structural flaw to be entirely reengineered.

This juxtaposition is of considerable importance when considering AI's role in technical pursuits. Optimization of constraints - adjusting parameters, substituting materials, exploring combinations - is precisely the kind of guided search that language models excel at. But structural redesign and reconception are fundamentally different activities. They require identifying that a framework's foundational assumptions are themselves the limiting factors and that the search space needs to be reinvented rather than fully explored.

Language models have access to vast and interdisciplinary knowledge, and it may be argued that the raw materials for such insights are to be found in their training data. However, the mere presence of disparate ideas and techniques cannot ensure that they will be joined together in a powerful reconceptualization. For example, a model

trained on every engine design before Watt would contain information about heat loss, thermal cycling, and the separation of functions in other mechanical contexts. But would an architecture based on statistical prediction surface such a radical recombination as a candidate solution? This would require overcoming the implicit tendencies in the distribution of the model's training data to generate statistically likely outputs.

Our results do not provide evidence of a model making such foundational jumps. The model explored the design space systematically and produced genuine moments of creative recombination. But nothing in its outputs suggested the capacity to offer radical restructuring that would overturn orthodoxy and bring new engineering ideas to the fore.

At the same time, the engine's history also reveals where AI's real value may reside. Alongside the radical reconsiderations, engines improved through the steady accumulation of smaller advances: more rigorous thermodynamic analysis, higher-temperature alloys, better lubricants, electronic fuel injection. These incremental, compounding refinements were collectively consequential and enduring. This is the domain most conducive to AI acceleration. At its core, the language model functions as a hyper-efficient search engine (Chauhan, 2025) - one that can systematically explore parameter spaces, test material configurations, and navigate constraint landscapes far more efficiently than manual intervention.

Our results demonstrate this: the system explored seven discrete designs while navigating constraints, decomposing abstract primitives into concrete parts lists, and identifying unproductive domain primitive combinations to avoid in subsequent attempts. This suggests a natural division of labor in which AI accelerates design space exploration and constraint navigation while human engineers supply the judgment, semantic understanding, and capacity for reconceptualization.

4.6 Architectural Limitations and Inductive Bias

Noting the limitations of language models for engineering design, would more compute translate to innovative breakthroughs? The history of model development provides a negative answer. Progress in machine learning has most often been driven by identifying new and robust inductive biases - the structural assumptions about the nature of the distribution and which learning representations are best suited to capture them (Battaglia et al., 2018).

This follows a well-known pattern. Recognizing that image pixels maintain spatial relationships with translational invariance led to convolutional neural networks (LeCun et al., 1998). Recognizing that contextual relationships vary in significance across sequential text led to transformers (Vaswani et al., 2017). In each case, capturing the right structural assumptions about the data produced dramatic performance gains.

The transformer architecture has demonstrated strong reasoning fluency, but there are grounds to question whether its inductive biases are sufficient for engineering innovation and scientific discovery. The engine breakthroughs we examined earlier were fundamentally acts of structural redesign. These require discontinuous leaps that have no immediate precursors. An autoregressive model operating on token sequences may be fundamentally limited in approximating this behavior.

What such an architecture would look like remains an open question. It would likely need to encode concepts like invariance, equivalence, and symmetry as representational primitives rather than emergent properties of token prediction. Whether this is possible - and whether it would be sufficient for the kind of structural redesign that drives paradigm-shifting breakthroughs - is a consequential yet unanswered question.

4.7 Directions for Architectural Improvement

Our linear pipeline was a deliberate choice. It provided a tractable, principled, and interpretable execution of the AI-orchestrated design process. But engineering design is fundamentally nonlinear. Engineers routinely move across levels of abstraction, refactor structures, and simultaneously balance competing constraints. Our system's unidirectional progression restricted this kind of fluid and free-form exploration.

A more flexible architecture would likely move toward multi-agent frameworks with a different allocation of responsibilities. Consider a custodian agent charged with maintaining the coherence of the base design, an innovator agent tasked with evolving components toward the synthesis target, and a mediating agent that adjudicates conflicts between them. Recent work on multi-agent debate has shown that assigning distinct roles and encouraging inter-agent deliberation improves reasoning outcomes (Du et al., 2023). This reflects a top-down approach to synthesis in contrast to our bottom-up construction.

Our own results point to the need for such flexibility. Design 3 was penalized for one of the more creative decisions in the experiment (selective abstraction of an energy mechanism) because the validator could only enforce rigid provenance rules. Design 1's semantic errors went unchallenged because no agent was charged with evaluating conceptual coherence. A system where agents can negotiate, challenge, and mediate rather than simply output a pass or fail outcome would better accommodate the kind of robust synthesis we desire.

Process parallelization offers another possible path for improvement. An updated system could run concurrent pipelines with different constraint hierarchies prioritizing cost, coherence, or manufacturability.

Whether either approach would yield stronger results remains to be tested, but the architectural limitations our system exposed suggest clear directions for further work.

4.8 Limitations

The primary limitations of this paper stem from its experimental scope and evaluation methodology. The adversarial pairing of a pistol shrimp and rock breaker was intentionally designed as an extreme stress test of AI synthesis capabilities. It is possible that the physics of these two systems are fundamentally irreconcilable and the task is intractable. The deep incompatibility of this pairing may preclude drawing insights about AI performance on routine engineering tasks where system compatibility is higher.

As a single case study, generalizability is limited. Results may not extend to other biomimetic pairings, non-adversarial synthesis tasks, or engineering domains beyond mechanical systems. The EDF framework and constraint architecture are specific to this experimental design.

Several evaluation constraints affect interpretation. No human engineers attempted the same synthesis task. This precluded baseline comparison of AI performance relative to expert capabilities. We employed no quantitative metrics for creativity or synthesis quality and instead relied on subjective assessments of creativity based on researcher judgment. Designs were neither physically prototyped nor validated by domain experts and their practical feasibility is unverified.

Methodologically, the system lacked access to external computational tools (e.g., CAD software, physics engines) that might have mitigated the quantitative reasoning failures observed. Additionally, constraint coverage was deliberately incomplete. We validated physics compliance and provenance accuracy but omitted real-world engineering factors including cost, manufacturability, safety, maintainability, and operational complexity.

The study examined a single model architecture (Claude Sonnet 4) with defined temperature settings (0.4 for synthesis, 0.1 for validation). Different model families, prompting strategies, temperature configurations, or computational budgets might yield substantially different outcomes. The token budget limit of 5 million may have prematurely terminated promising architectural explorations.

Despite these limitations, the framework's ability to generate interpretable failure modes and surface identifiable moments of creative reasoning suggests that constrained synthesis pipelines may be a productive framework for studying AI's engineering capabilities.

5. Conclusion

We designed a hierarchical, constrained multi-agent synthesis framework to test whether a language model could synthesize a pistol shrimp and hydraulic rock breaker into a coherent hybrid mechanical construction. The deliberate adversarial pairing ensured failure but created conditions for studying how language models reason under constraints - where they degrade, where they show evidence of

coherence and novelty, and what these patterns reveal about the feasibility of AI-orchestrated engineering design.

The system produced seven discrete designs of varying technical complexity. We identified design choices reflecting genuine creative problem-solving – energy mechanism abstraction across domains, spontaneous invention of notation, independent convergence on established engineering strategies. But the system also exposed persistent weaknesses: spurious rationalizations masking semantic misunderstanding, difficulty sustaining multi-step inference, and failure to maintain consistency across long-range dependencies. Overall however the system demonstrated competence in exploring the design space and surfacing novel configurations.

Language models are well-suited to optimizing parameters and navigating constraints and this positions them well for accelerating incremental engineering tasks. But groundbreaking advances require radical restructuring - discontinuous leaps with no immediate precursors. An autoregressive model operating on token sequences may be fundamentally limited in approximating this kind of reasoning. Nothing in our results suggests that current AI architectures can transcend this limitation.

Advancing AI's role in technical disciplines requires both architectural innovation and effective human-AI collaboration. Architecturally, new inductive biases encoding concepts like invariance, symmetry, and structural equivalence would better equip models for the reasoning demanded by engineering and scientific discovery. In terms of collaboration, our results point toward a more targeted division of labor. This calls for AI systems accelerating design space exploration, component selection, and constraint navigation, and human engineers supplying the domain judgment, semantic understanding, and capacity for reconceptualization that models cannot yet provide.

6. Data and Code Availability

The complete source specifications (pistol shrimp and hydraulic rock breaker YAML files), constraint configurations (physics.yaml, structural.yaml, mechanism_prerequisites.yaml, edf_codes.yaml), system prompts for all agents, the full EDF framework document, orchestration engine code, raw JSON outputs for all seven design trajectories, and research dossiers documenting the source literature for both systems are available at <https://github.com/alvco/AI-Engineering-Design>.

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